Autoreason: Autoresearch for Subjective Domains

Nous Research

https://github.com/NousResearch/autoreason

1 Problem

LLMs used for iterative refinement exhibit compounding failure modes: **sycophancy** (strengthens whatever you hand it), **overcriticism** (always finds problems even when work is sound), **overcompromise** (hedges everything when synthesizing), **authorship bias** (defends its own prior choices), **scope drift** (adds complexity each iteration), and **context collapse** (output diverges from original intent with no reversion mechanism). The output is shaped more by how you prompt than by what is actually better.

2 Method

Autoreason extends autoresearch to subjective domains by constructing a fitness function from independent blind evaluation—the same way science uses peer review where math can use proofs.

Every role is a fresh, isolated agent with no shared context. The loop:

1. **Version A** — the incumbent (unchanged)
2. **Strawman** — fresh agent attacks A, finds problems only
3. **Version B** — fresh agent revises A using valid criticisms
4. **Version AB** — fresh agent synthesizes best of A and B
5. **Judge panel** — 3 blind judges rank A, B, AB against the original task prompt (randomized labels, Borda count, incumbent wins ties)

Winner becomes new A. Loop repeats until A survives 2 consecutive passes.

A represents conservatism, **B** represents adversarial editing, **AB** approximates objectivity. The judge panel decides which framing produced the best result, with no knowledge of which is which.

3 Results

3.1 Convergence Trajectory

We ran 26 passes on a go-to-market strategy task (Claude Sonnet 4, 3-judge panel). Figure 1 shows the full trajectory.

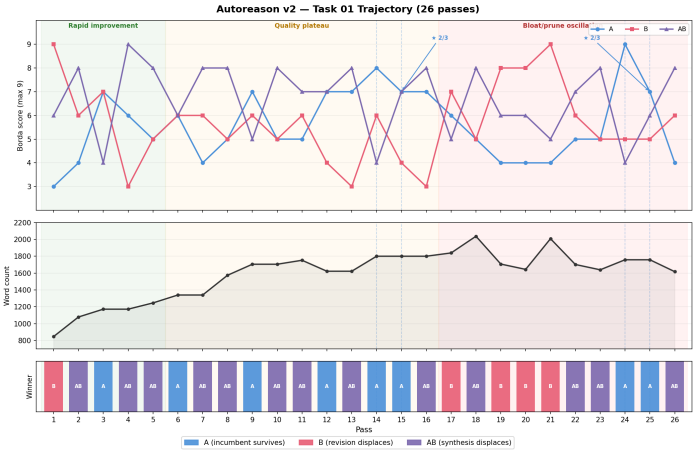

Figure 1: **Autoreason v2 trajectory (26 passes).** Top: Borda scores per pass. Middle: word count showing bloat/prune oscillation. Bottom: winner per pass. Dashed lines mark 2-consecutive-A-win points (passes 14–15 and 24–25).

The loop reached 2 consecutive incumbent wins at pass 15. Key dynamics: early passes show genuine improvement (pass 1: unanimous B win, Borda 9-3-6). Fresh agents prevent authorship bias—Author B re-emerged at passes 17–21 after 15 passes of irrelevance. On ambiguous tasks, the synthesizer (AB) adds complexity while Author B prunes it, creating a bloat/prune oscillation (847 → 1800 → 1644 → 1758 words).

3.2 Qualitative Improvement

The initial output is a generic startup playbook with round-number targets and no validation. After 14 passes, the autoreason output contains quantified pain ($15K/incident × 6/year), team-based pricing matching buying motions ($1,499/mo vs $49/user), validation from 50+ interviews with 75% pilot success, competitive positioning against named tools, and unit economics (CAC $2K, LTV $54K). Unrealistic revenue targets were killed ($100K → $25K MRR).

3.3 5-Way Baseline Comparison

All methods start from the same initial output (847 words), 15 passes, same model. 7-judge blind panel with randomized labels, Borda count scoring.

Method	Borda	1st	Words
Autoreason	**35/35**	**7/7**	1,800
Conservative	21	0	862
"Improve this"	18	0	2,116
Harsh critic	18	0	1,961
Critique & revise	13	0	2,507

Table 1: **7-judge blind panel results.** Autoreason: perfect score, all first-place votes. Conservative (barely changed) outranks all iterative baselines. Critique-and-revise, the standard approach, came last.

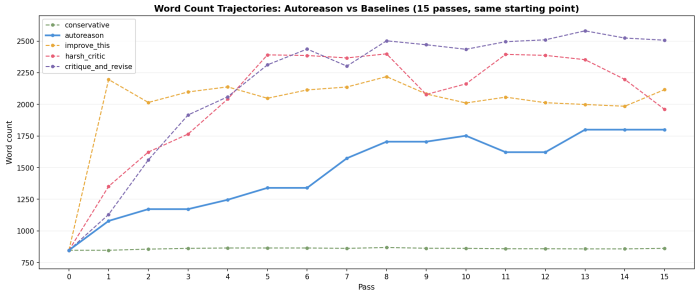

Figure 2: **Word count trajectories.** Conservative flatlines, autoreason stabilizes, all iterative baselines bloat. Each method's structural bias is visible in the trajectory shape.

Doing almost nothing (conservative) produced better output than 15 passes of "improve this," "harsh critic," or "critique and revise." Iterating without structural protection is worse than not iterating.

3.4 Convergence Threshold

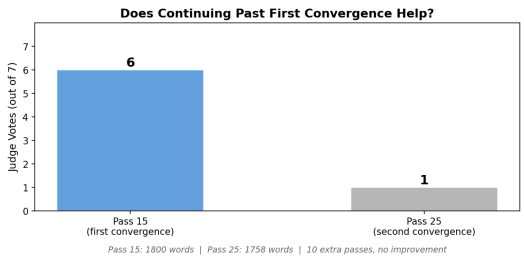

Figure 3: **Does continuing past first convergence help?** Pass 15 (1800 words) vs pass 25 (1758 words), 7-judge blind panel: 6–1 for pass 15. First convergence is the quality ceiling.

The first convergence point is the quality ceiling. Ten additional passes degraded output—the bloat/prune oscillation churned without improving.

3.5 Judges Need a Baseline

Two panels compared autoreason vs critique-and-revise directly. Without the initial output shown: 3–2. With it: 7–0. Judges need a reference point to distinguish improvement from drift.

4 Prior Work

Earlier experiments (∼1,800 LLM calls) informed the v2 design. Key findings: severe positional bias in non-blind evaluation (eliminated by randomized labels), single-judge noise (fixed by 3-judge panel), shared-context contamination where the same agent rubber-stamped its own revisions (fixed by fresh isolated agents per role).

5 Related Work

Autoresearch (Karpathy) uses val_bpb as an objective fitness function for autonomous experiments. Autoreason extends this to domains without objective metrics. **SlopCodeBench** (Orlanski et al., 2026) shows agents produce progressively worse code each iteration—prompting doesn't fix it. **ACE** (Zhang et al., ICLR 2026) documents context collapse in iterative rewriting. **LLM Council** (Karpathy) uses multi-model panels for single-turn evaluation; autoreason adds iteration with role separation.

6 Open Questions

Does autoreason improve outcomes where metrics exist but are insufficient (code that passes tests but degrades)? Do mixed-model judge panels converge faster? Does the loop reach the same optimum across independent runs?

Code, data, and artifacts: `https://github.com/NousResearch/autoreason`